Reimagining Mapping: Food Insecurity in New York State

Sustainable Urban Environments Capstone

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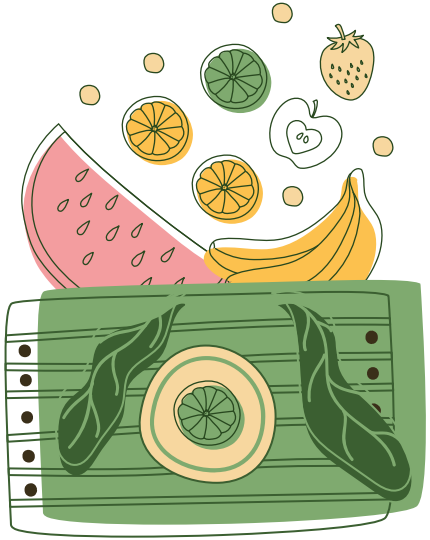


01

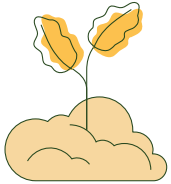
INTRODUCING FOOD INSECURITY

WHAT IS FOOD INSECURITY?

"The **limited or uncertain availability of nutritionally adequate and safe foods**, or limited or uncertain ability to acquire acceptable foods in socially acceptable ways."

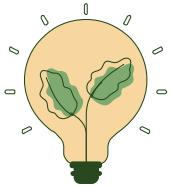
- USDA ERS

WHY PROPOSE A NEW FRAMEWORK?



PROBLEM

Limited mapping resources on food insecurity **available** for **local communities**



SOLUTION

Create an **all-inclusive** and **accessible** resource that can **inform and help** users

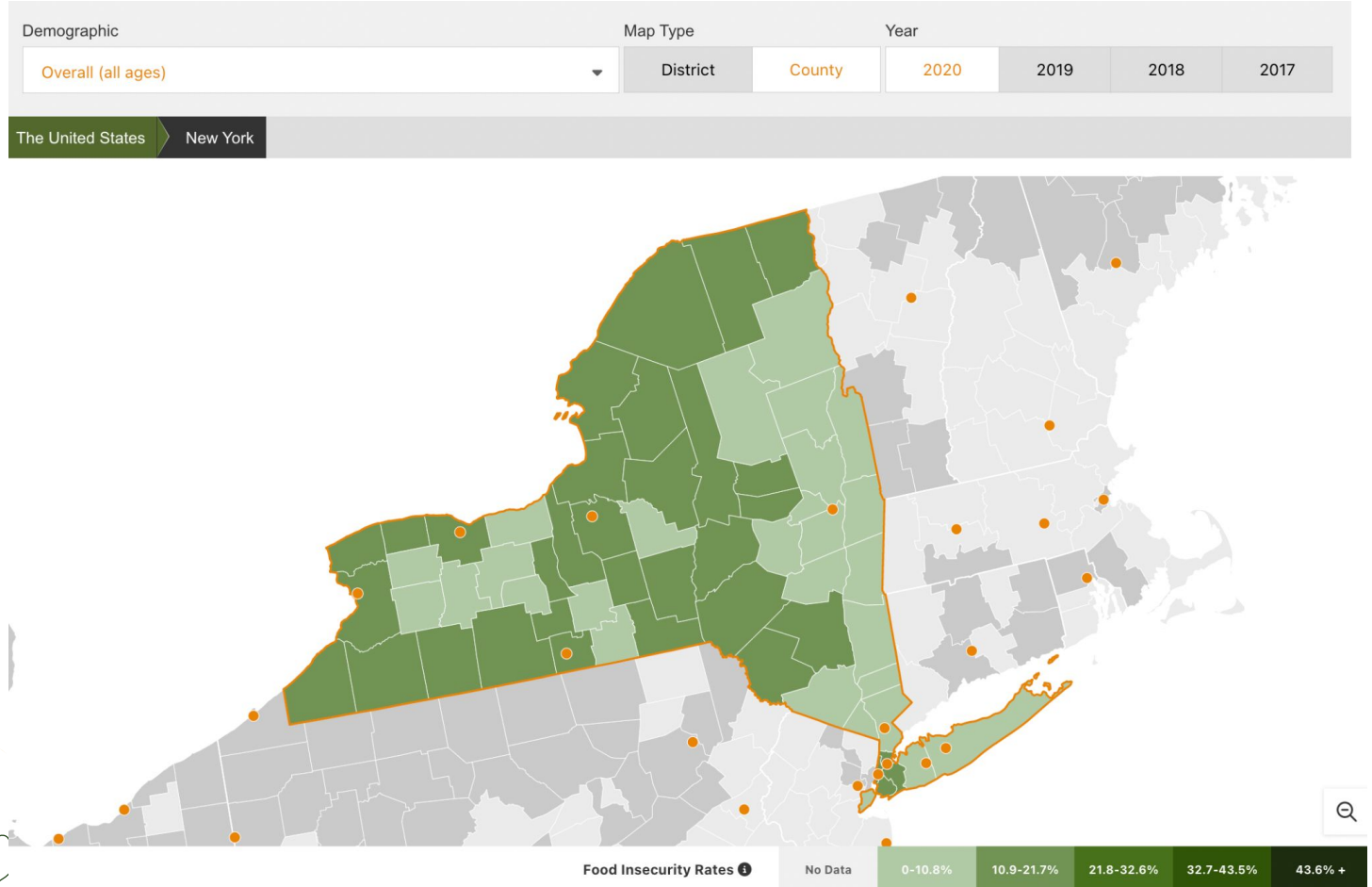
WHY DOES IT MATTER?

- Creating accessibility for all
- Promoting access to food benefits
- Value adding for minorities
- Comprehensive approach to understanding food insecurity

DON'T CHILL BELOW 55°
TAMPAX

FOOD INSECURITY IN THE US

FOOD INSECURITY IN NYS



02

EXAMINING EXISTING MAPPING TOOLS

LIMITATIONS OF CURRENT FOOD INSECURITY MODELS

- Indigenous populations are still underrepresented
- Lacking social justice conversation
- Resources are dispersed over various platforms

Food Insecurity among Overall (all ages) Population in the United States

Click or touch on the map, or use the search bar to start exploring

Search

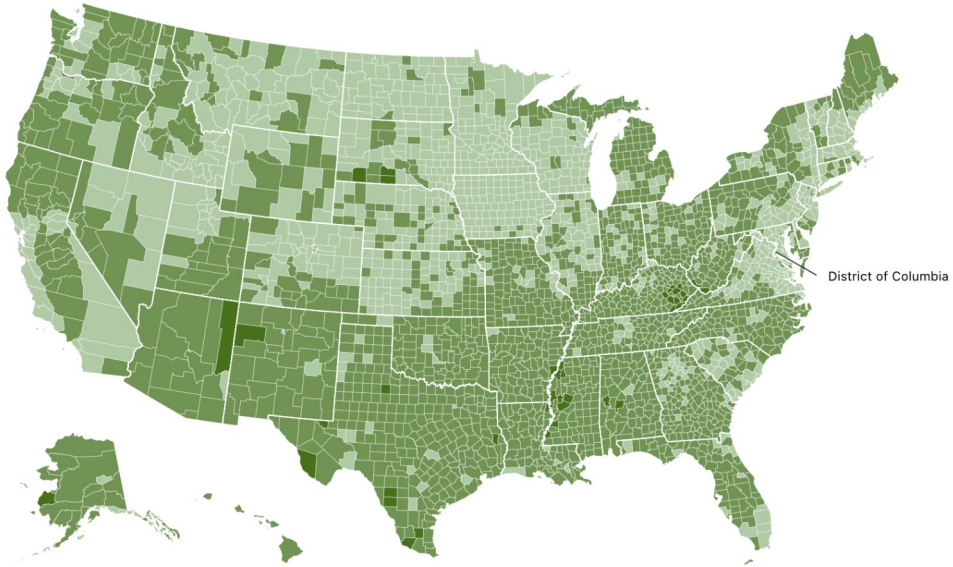
State, County, or Food Bank

Demographic: Overall (all ages)

Map Type: District | County

Year: 2020 | 2019 | 2018 | 2017

The United States



Food Insecurity Rates: No Data | 0-10.8% | 10.9-21.7% | 21.8-32.6% | 32.7-43.5% | 43.6% +

MAP THE MEAL GAP

FEEDING AMERICA

Data Used

- Current Population Survey (CPS)
- American Community Survey (ACS)
- Bureau of Labor Statistics (BLS)

AGGREGATING FOOD INSECURITY

STATE:

$$FI_{st} = \alpha + \beta_{UN}UN_{st} + \beta_{POV}POV_{st} + \beta_{MI}MI_{st} + \beta_{HISP}HISP_{st} + \beta_{BLACK}BLACK_{st} + \beta_{OWN}OWN_{st} + \beta_{DSBL}DSBL_{st} + \mu_t + \upsilon_s + \varepsilon_{st} \quad (1)$$

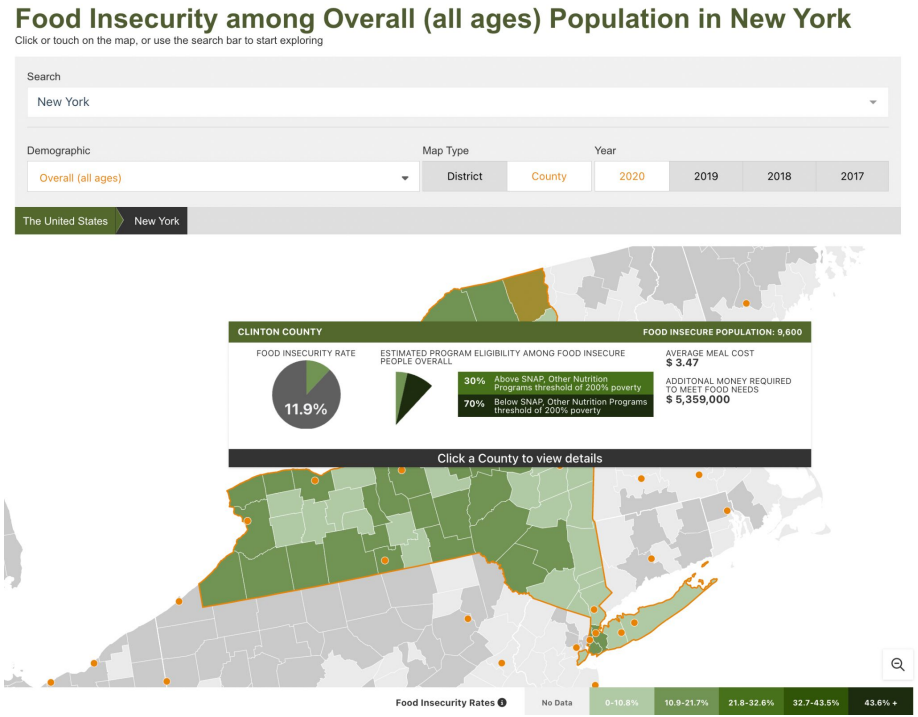
COUNTY:

$$FI^*_c = \hat{\alpha} + \widehat{\beta_{UN}}UN_c + \widehat{\beta_{POV}}POV_c + \widehat{\beta_{MI}}MI_c + \widehat{\beta_{HISP}}HISP_c + \widehat{\beta_{BLACK}}BLACK_c + \widehat{\beta_{OWN}}OWN_C + \widehat{\beta_{DSBL}}DSBL_C + \widehat{\mu_{2018}} + \hat{v}_s \quad (2)$$

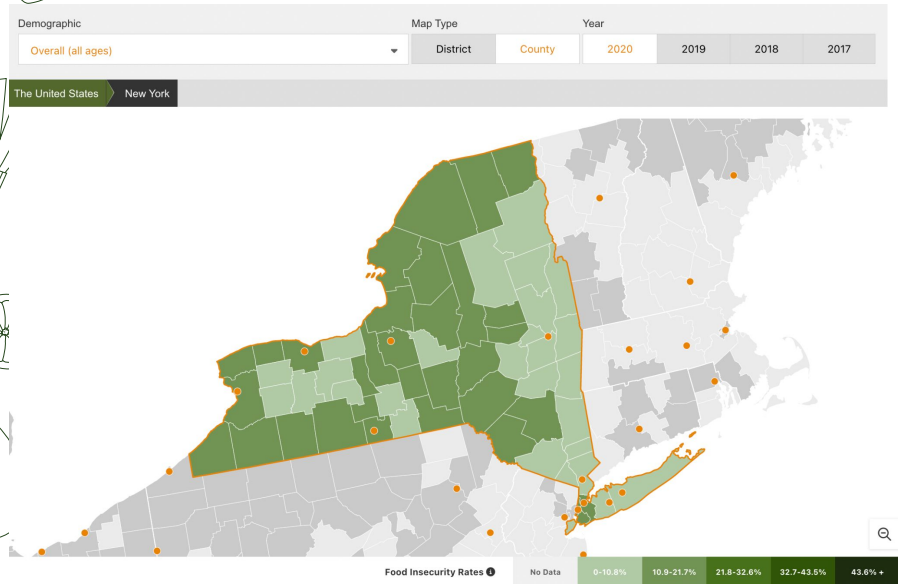
STRENGTHS & LIMITATIONS

STRENGTHS

- **Interactive**, **easy to use** and **informative**
- Uses **color gradients** to differentiate severity of food insecurity
- Data spanning multiple years to **show changes and trends**
- Considers racial and economic disparities in model
- Data from multiple sources



STRENGTHS & LIMITATIONS



LIMITATIONS

- Visual differentiation on local level
- **Not all BIPOC populations** are considered
  - Only **Black, Latino, and White non-Hispanic** populations are considered in the aggregation

Food Insecurity and Federal Nutrition Program Participation

A county-level reporting of participation in SNAP, WIC, Summer Food Service Program, and CACFP Afterschool Meals.

Click on a County to view participation details.

Chautauqua County

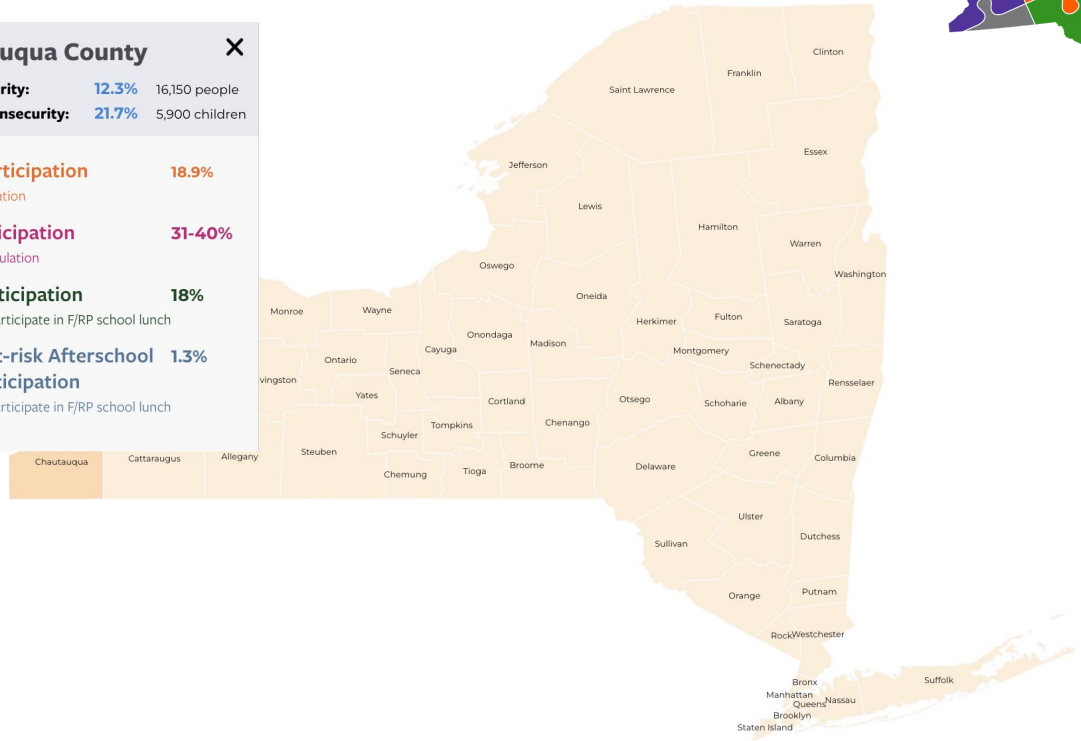
Food Insecurity:	12.3%	16,150 people
Child Food Insecurity:	21.7%	5,900 children

SNAP Participation 18.9%
of total population

WIC Participation 31-40%
of eligible population

SFSP Participation 18%
of kids who participate in F/RP school lunch

CACFP At-risk Afterschool Meal Participation 1.3%
of kids who participate in F/RP school lunch



HUNGER SOLUTIONS NEW YORK
UNITING POLICIES AND PROGRAMS TO END HUNGER

Data Used

- Map the Meal Gap (MMG)
- Food Research & Action Center (FRAC)
- NYS Department of Health
- HSNY Summer Food Service Program & At-risk Afterschool Meals

STRENGTHS & LIMITATIONS

STRENGTHS

- Interactive and easy to navigate
- Combines data from various sources
  - Show multiple program participation rates
- Focuses on county-level data

LIMITATIONS

- Uses data from different years
- No visual differentiation by county

03

BUILDING A FOOD INSECURITY FRAMEWORK

WHY PROPOSE A NEW FRAMEWORK?

CURRENT MODELS

- Challenging to accurately quantify food insecurity
- Existing models impact how we perceive and prioritize resources to address the issue

PURPOSE OF THE TOOL

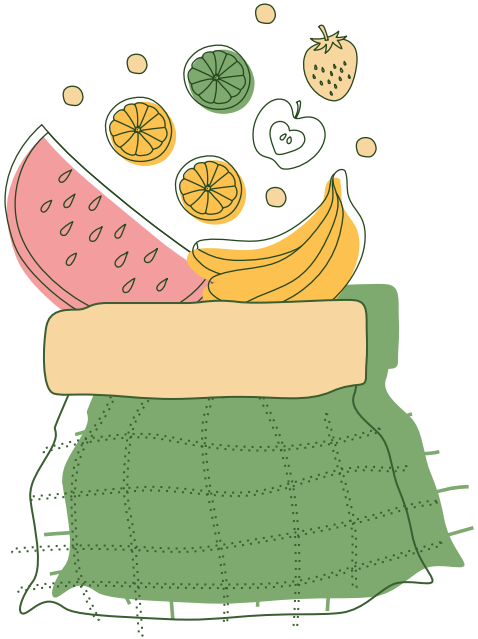
- Inform laypeople
- Inform policy makers
- Advocate for change in food policy

HOW TO IMPROVE CURRENT MODELS

- Increasing available information and data
- Considering more nuanced circumstances
- Impact of resources

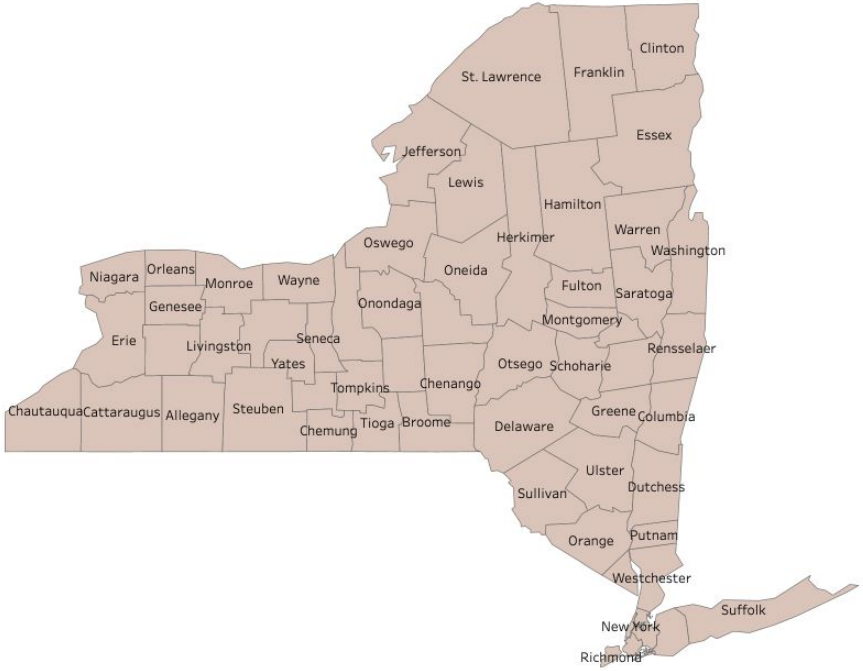
04

MODELING FOOD INSECURITY



Welcome to New York State Food Map

Learn more about food security in all New York State Counties through our interactive map. Click to play.



MY PROPOSAL

NYS FOOD MAP

About Us | Get the Data | FAQ | Log In

Filter by Keywords

Search by County

- Select All Counties
- Alabany
- Allegany
- Bronx
- Broome

Demographics

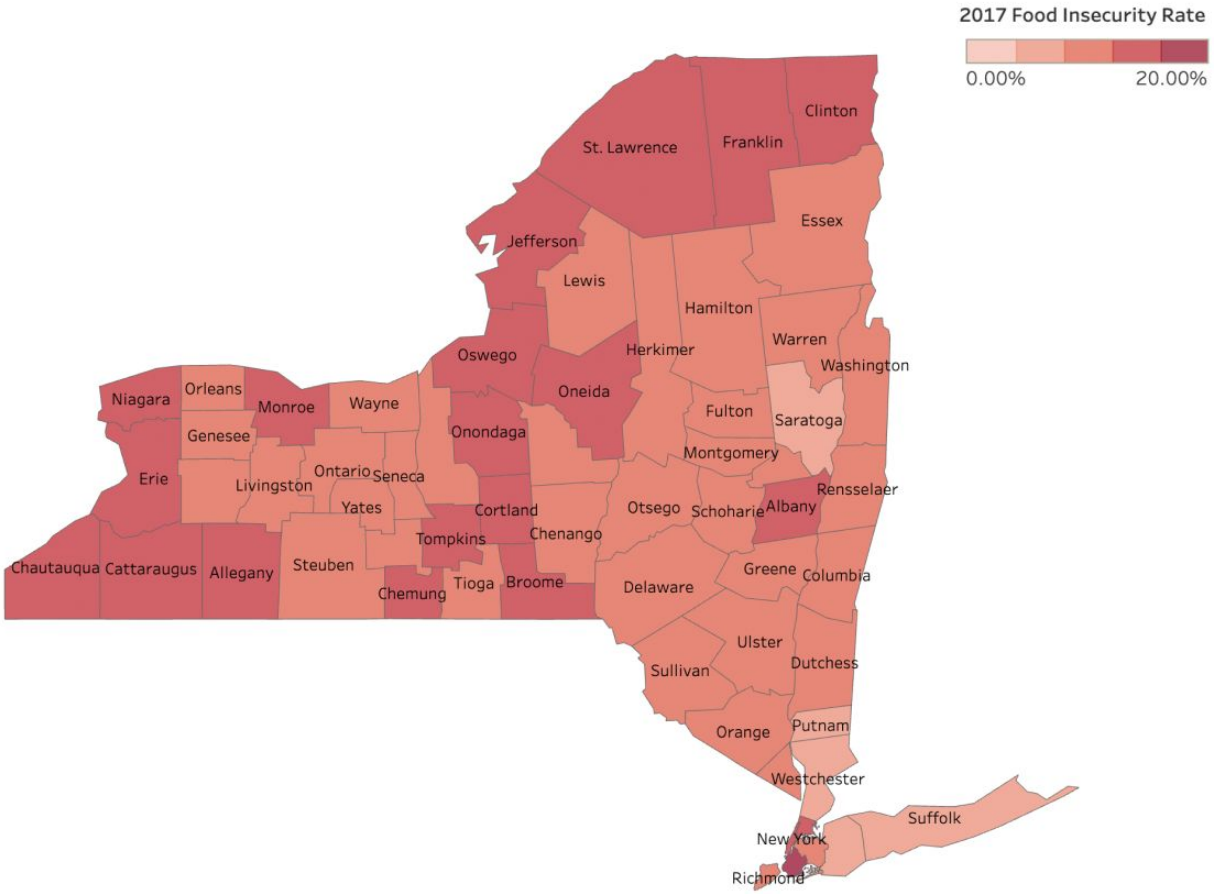
Food Insecurity

Nutrition Assistance

Average Meal Cost

Emergency Food

Income



Auditory Assistance

Language

Settings

Help center

2017 Projected

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Food Insecurity

- Food Insecure Population
- Food Insecurity Rate

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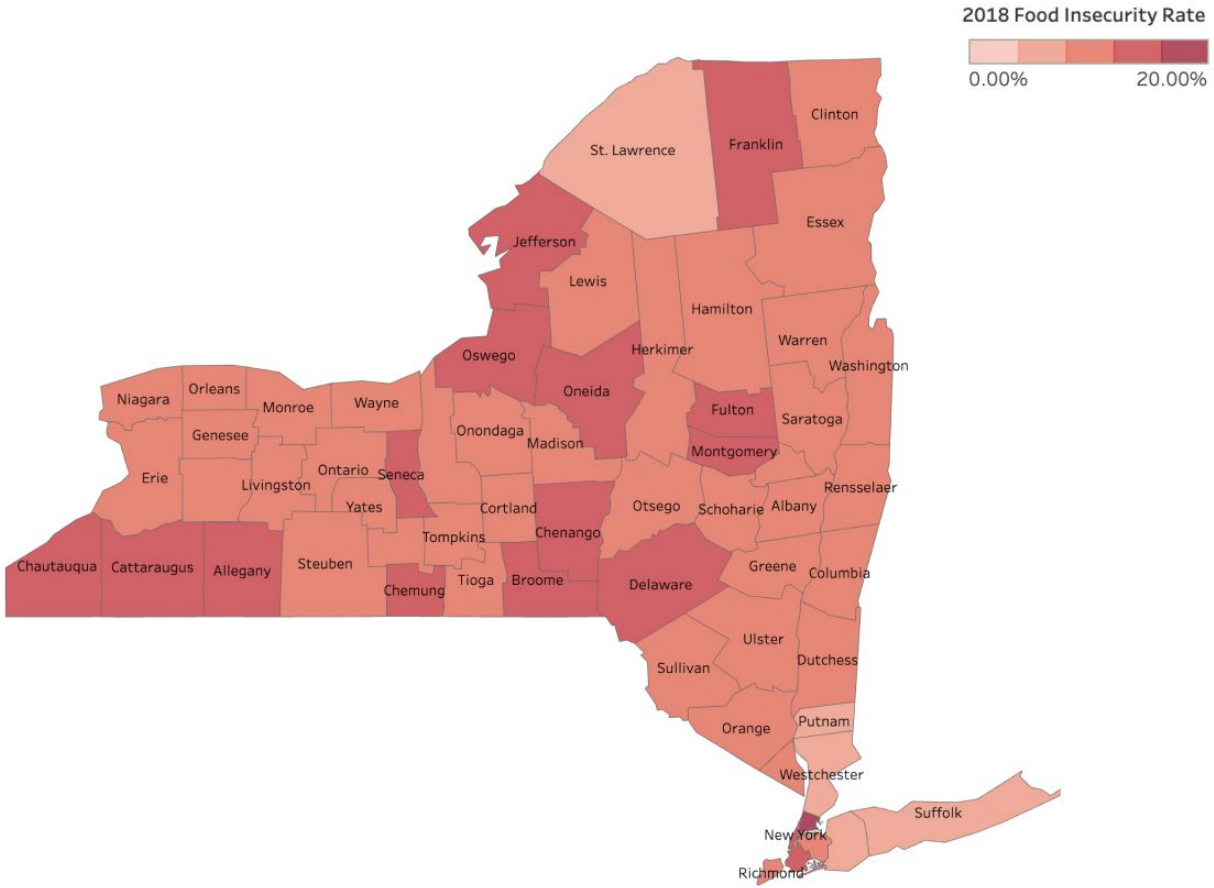

2018 Projected

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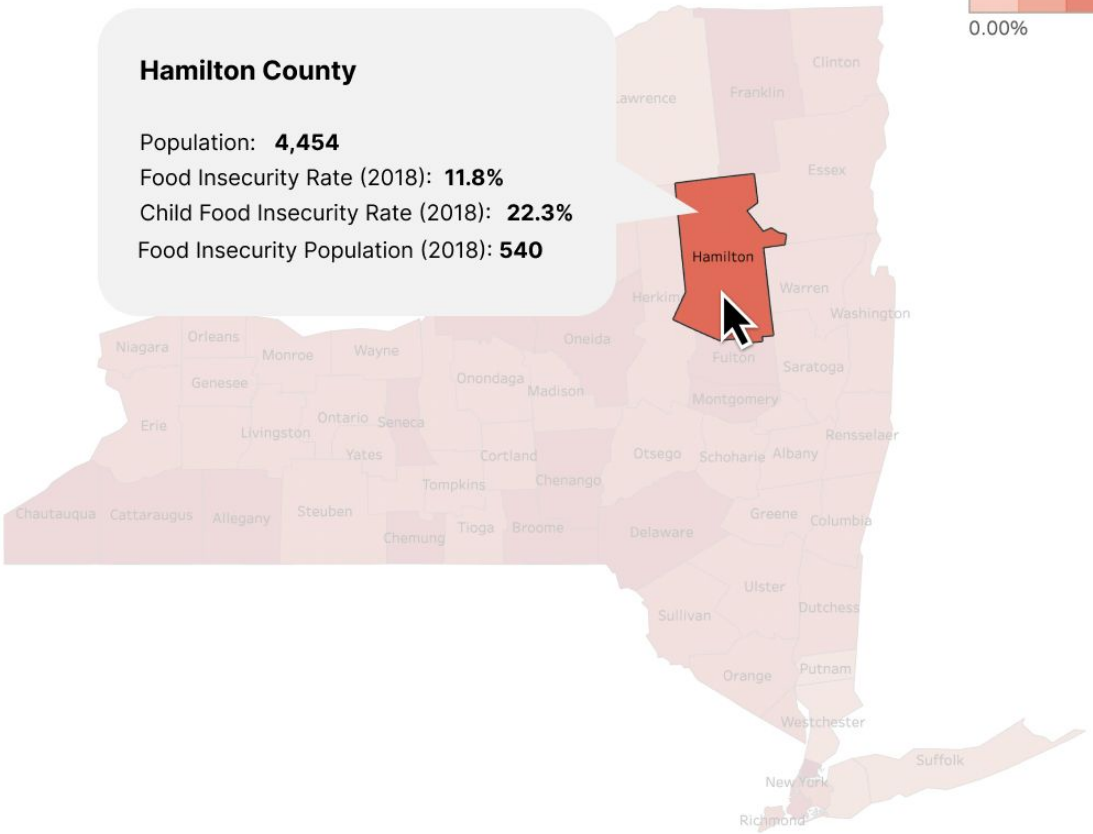
2018 Food Insecurity Rate

0.00% 20.00%



Hamilton County

Population: **4,454**

Food Insecurity Rate (2018): **11.8%**

Child Food Insecurity Rate (2018): **22.3%**

Food Insecurity Population (2018): **540**

2018 Projected

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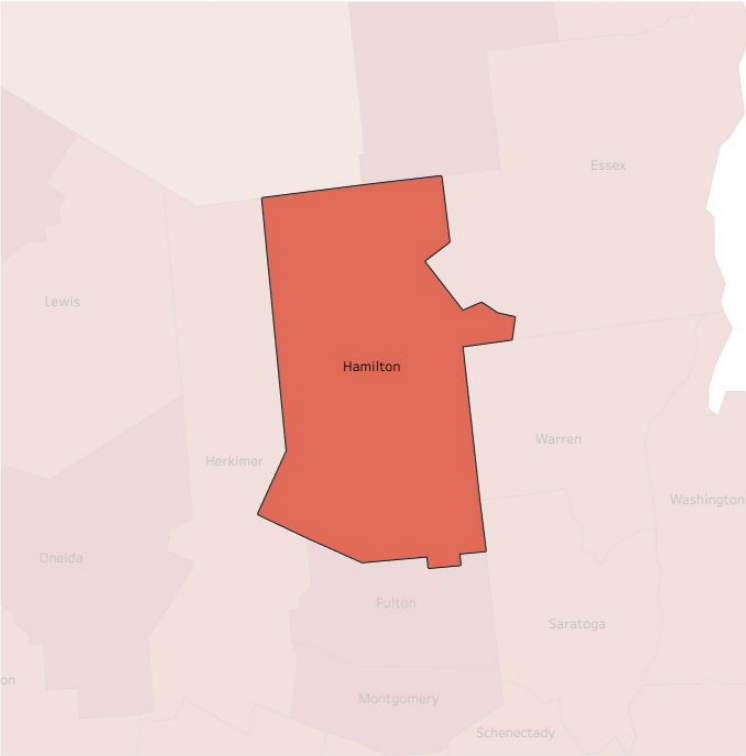

Hamilton County

Population: **4,454**

Food Insecurity Rate (2018): **11.8%**

Number of Food Insecure Persons (2018): **540**

Child Food Insecurity Rate (2018): **22.3%**

Number of Food Insecure Children (2018): **140**

Find out about local food resources available to you
https://regionalfoodbank.net/agency-list-by-county/hamilton/
https://www.hamiltonhelps.org/what-you-can-give

Learn more about Hamilton County, NY
https://www.hamiltoncounty.com/

Data retrieved from
US Census 2020 ACS Demographics
Map the Meal Gap 2020

MY PROPOSAL

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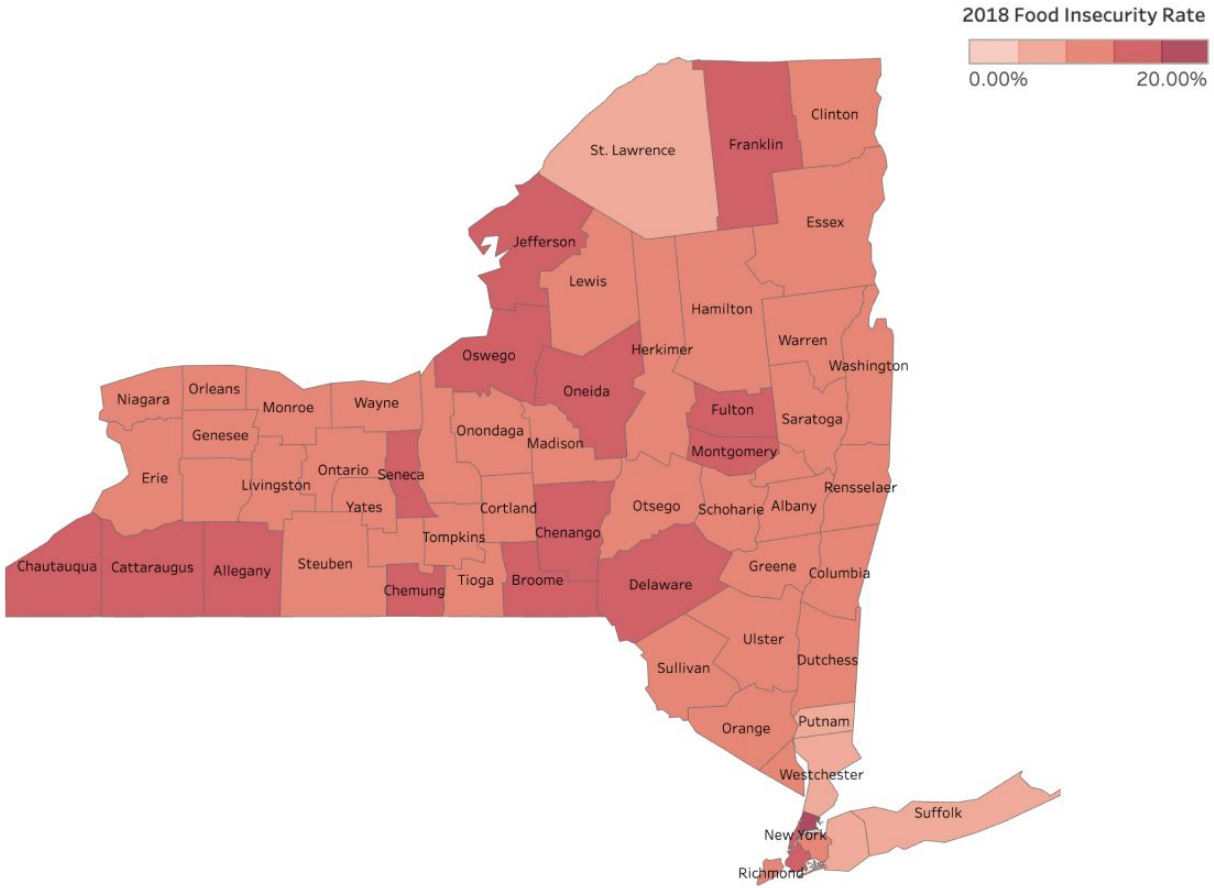

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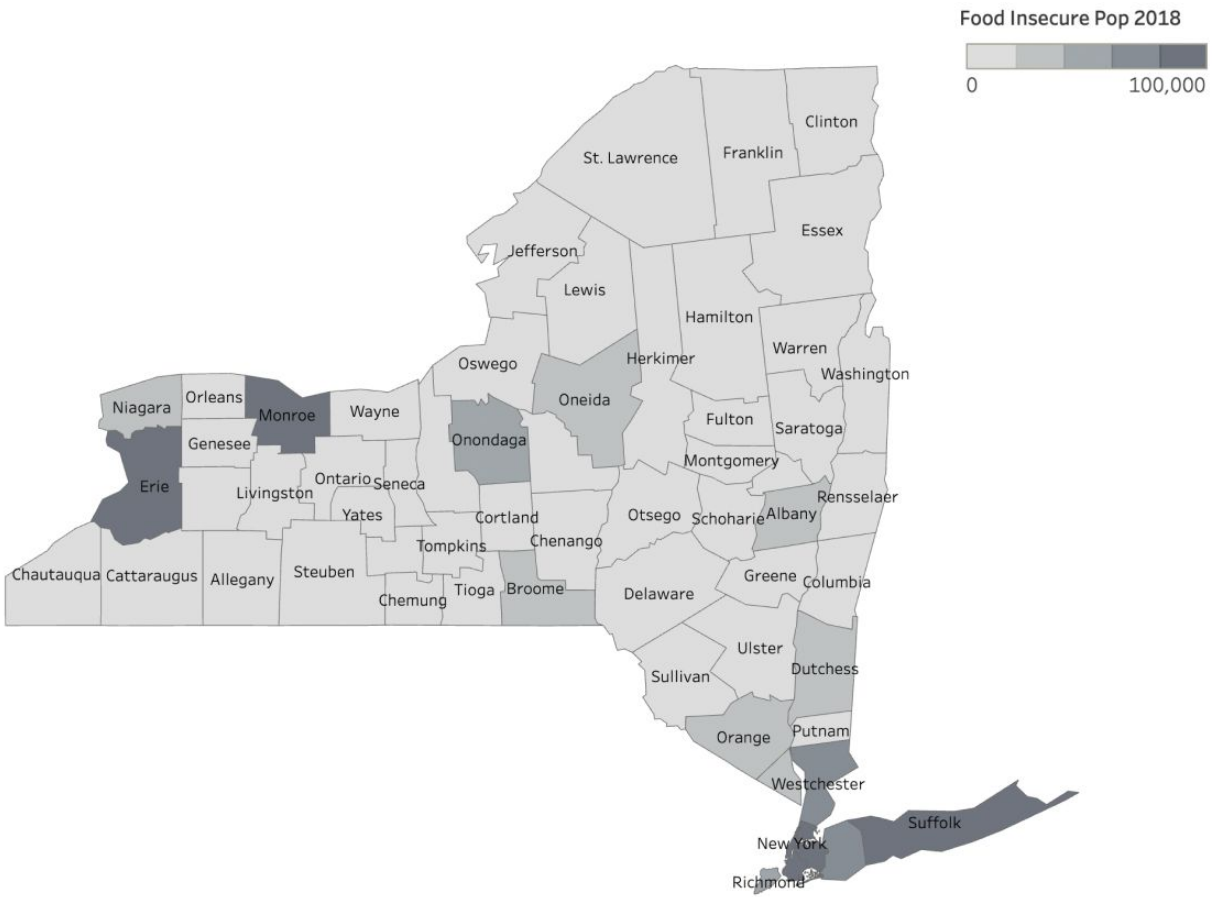

2018 Projected

MY PROPOSAL

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THANKS!

Do you have any questions?

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